

Avaliação III - Redes de Computadores - 19/07/2023 VALOR: 8,0 (peso 2,0)

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(1,0) 1 - Uma organização possui uma rede de classe C 200.1.1.0 e deseja formar sub-redes para quatro departamentos, com os 145 hosts divididos da seguinte forma:
A) 72 hosts; B) 35 hosts; C) 20 hosts; D) 18 hosts;

É possível segmentar a rede, criando 4 sub-redes que atendam aos requisitos acima?
Em caso afirmativo, para cada sub-rede informe: o endereço da sub-rede, endereço de broadcast e máscara de sub-rede.

(1,0) 2 - Sobre o CSMA/CD, indique qual será o intervalo no qual a variável K estará contida após ter havido 20 colisões.

(1,0) 3 - Considere o seguinte endereço IPv6 e marque a alternativa que possui um endereço equivalente no formato mais abreviado possível.

Endereço original: 2001:0db8:0000:130F:0000:0000:087C:140b

- a) 21:db8::13F::87C:14b
- b) 2001:0db8::130F::087C:140b
- c) 2001:0db8:0:130F:0:0:087C:140b
- d) 2001:0db8::130F:0:0:087C:140b
- ☒ e) 2001:0db8:0:130F::087C:140b

(1,0) 4 - A partir do curso de IPv6 aprendemos que a IANA (Internet Assigned Numbers Authority), reservou para a Internet endereços IPv6 roteáveis na faixa 2000::/3. Com base nessa informação, responda quais são o menor e o maior endereço IPv6 possíveis de serem obtidos dentro dessa faixa.

(1,0) 5 - Considere que um datagrama original (com tamanho de 4500 Bytes) precisou ser fragmentado em um roteador cujo tamanho da MTU é de 1412. Preencha os campos com os valores corretos nos datagramas após a fragmentação.

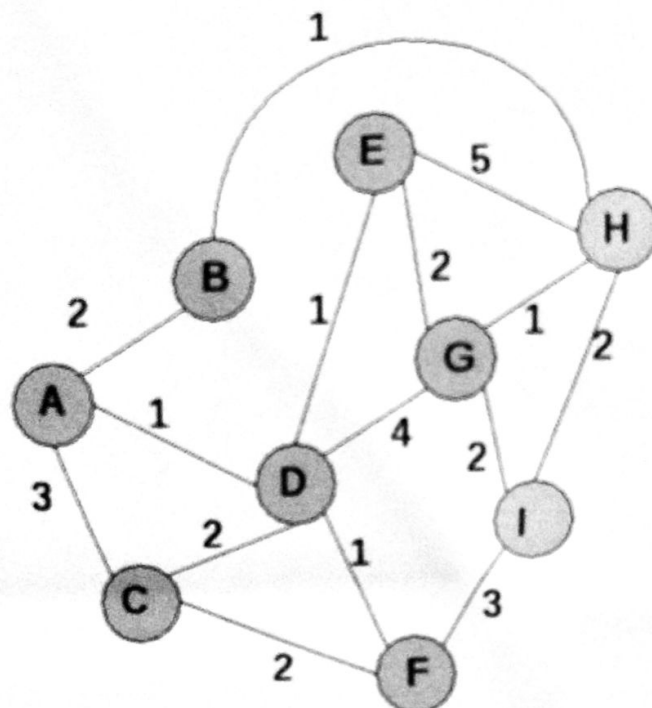
Datagramas após a fragmentação:

Length = 1412	ID=X	fragflag = 1	offset = 0	dados
Length = 1412	ID=X	fragflag = 1	offset = 1412	dados
Length = 1412	ID=X	fragflag = 1	offset = 2824	dados
Length = 324	ID=X	fragflag = 0	offset = 522	dados

(1,0) 6 - Explique como o protocolo SSL previne o ataque de alteração do campo de número de sequência no cabeçalho do protocolo TCP. *Detalhar*

(1,0) 7 - Explique a função do campo ESP Trailer no cabeçalho do IPSEC.

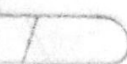
(1,0) 8 – Sobre o algoritmo de roteamento de estado do enlace, complete a tabela a seguir (semelhante à trabalhada em sala), mostrando como seriam os passos da construção da mesma com esse algoritmo. A tabela a ser construída é com relação ao roteador G. Os valores das arestas no grafo representam o custo para se enviar um pacote.



N'	D(a), p(a)	D(b), p(b)	D(c), p(c)	D(d), p(d)	D(e), p(e)	D(f), p(f)	D(g), p(g)	D(h), p(h)	D(i), p(i)
G	∞	∞	∞	4, g	2, g	∞	-	5, g	2, g
GH	∞	2, h	∞	4, g	2, g	∞	-	-	2, g
GHE	∞	2, h	∞	3, e	-	∞	-	-	2, g
GHEI	∞	2, h	∞	3, e	-	5, i	-	-	-
GHEID	4, b	-	∞	3, e	-	5, i	-	-	-
GHEIDB	4, b	-	5, d	-	-	4, d	-	-	-
GHEIDBA	-	-	5, d	-	-	4, d	-	-	-
GHEIDBAF	-	-	5, d	-	-	-	-	-	-
GHEIDBAFI	-	-	-	-	-	-	-	-	-

Boa Avaliação!

Uls: Coração Flávio



1. 200.1.1.0/24 ^{classe c}
7 bits = host
 $2^7 = 128 - 2 = 126$

1º rede = 200.1.1.0/25
host = 200.1.1.127
máscara = 255.255.255.128

2º rede = 200.1.1.128/26
host = 200.1.1.191
máscara = 255.255.255.192

3º rede = 200.1.1.192/27
host = 200.1.1.223
máscara = 255.255.255.224

4º rede = 200.1.1.224/27
host = 200.1.1.255
máscara = 255.255.255.224

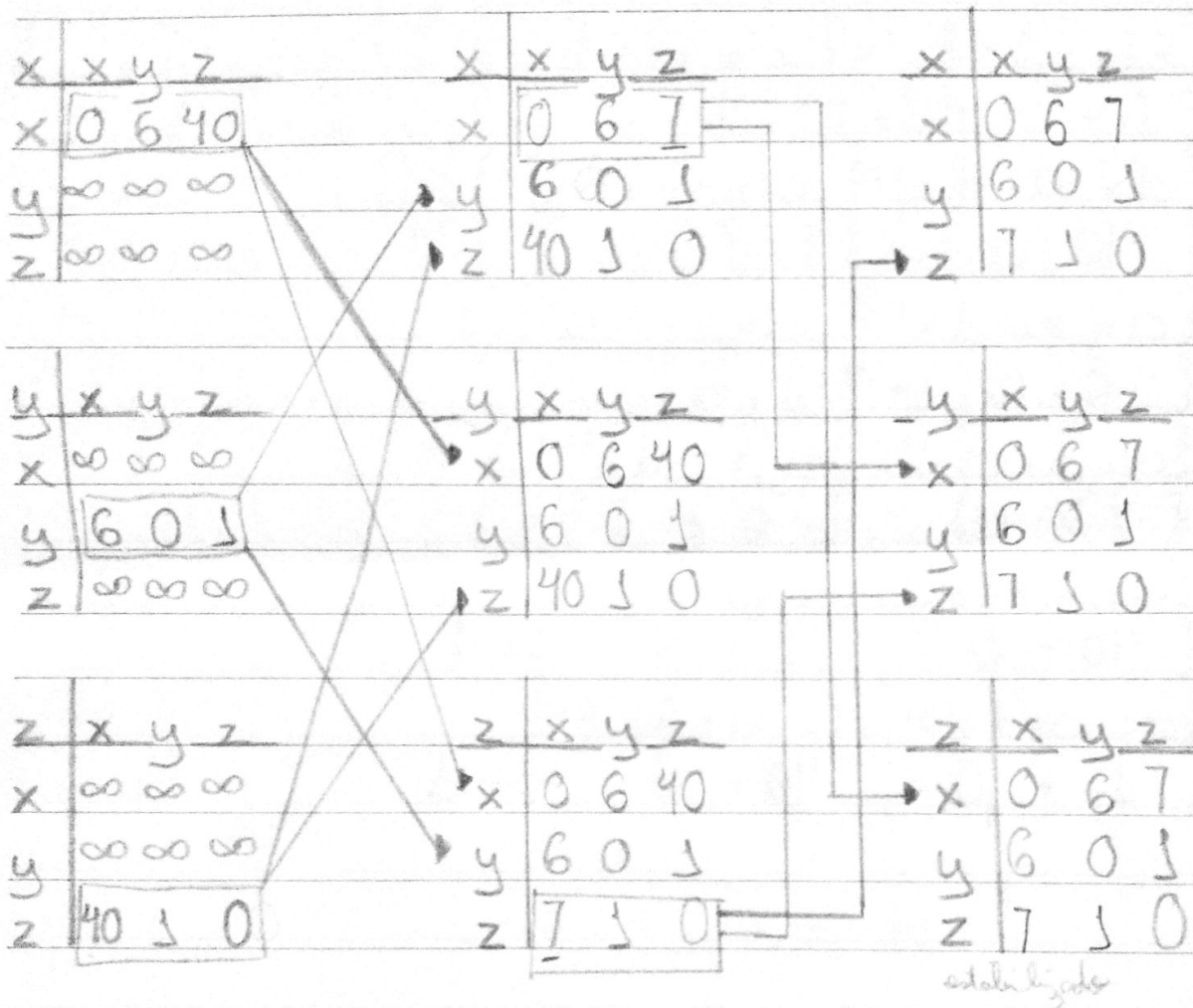
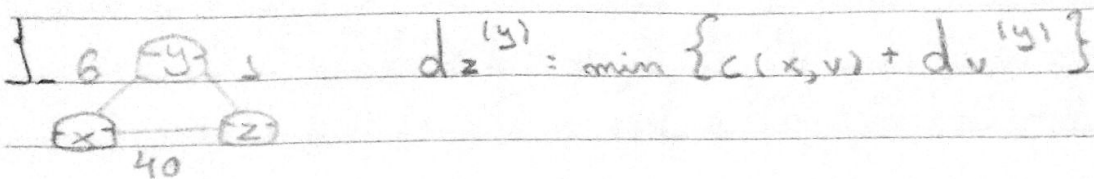
Questão 4. 2000::13 até
3FFF FFFF FFFF FFFF FFFF FFFF FFFF FFFF

Questão 6. Para um usuário mal intencionado alterar o campo do número de sequência, o SSL ainda o possuirá internamente, através dos mecanismos de criptografia, autenticação e verificação de integridade.

Questão 7. O processo de autenticação de integridade do campo ESP Trailer envolve o uso de uma função de hash ou HMAC. O remetente, ao criar o pacote aplica essa função ao conteúdo do payload criptografado e ao cabeçalho ESP, inserindo o valor resultante do cálculo ao campo ESP Trailer. Ao receber o pacote, o receptor realiza o mesmo cálculo, em seguida compara com o valor presente no campo ESP Trailer do pacote. Se ambos os valores coincidirem indica que a integridade está preservada. No caso contrário sugere que o pacote pode estar corrompido.

Guilherme Moraes Costa 2020.1.08.012

Lista de Exercícios - Vetor Distância



$$d_x^{(y)} = \min \{ c(x, y) + d_y^{(y)}, c(x, z) + d_z^{(y)} \}$$

$$= \min \{ 6 + 0, 40 + 1 \} = 6$$

$$d_x^{(z)} = \min \{ c(x, z) + d_z^{(z)}, c(x, y) + d_y^{(z)} \}$$

$$= \min \{ 40 + 0, 6 + 1 \} = 7$$

$$d_y^{(x)} = \min \{ c(y, x) + d_x^{(x)}, c(y, z) + d_z^{(x)} \}$$

$$= \min \{ 6 + 0, 1 + 40 \} = 6$$

$$d_y^{(z)} = \min \{ c(y, z) + d_z^{(z)}, c(y, x) + d_x^{(z)} \}$$

$$= \min \{ 1 + 0, 6 + 40 \} = 1$$

$$dz^{(x)} = \min \{c(z, x) + dx^{(x)}, c(z, y) + dy^{(x)}\}$$

$$= \min \{40 + 0, 1 + 6\} = 7$$

$$dz^{(y)} = \min \{c(z, y) + dy^{(y)}, c(z, x) + dx^{(y)}\}$$

$$= \min \{1 + 0, 40 + 6\} = 1$$

$$dx^{(y)} = \min \{c(x, y) + dy^{(y)}, c(x, z) + dz^{(y)}\}$$

$$= \min \{6 + 0, 40 + 1\} = 6$$

$$dx^{(z)} = \min \{c(x, z) + dz^{(z)}, c(x, y) + dy^{(z)}\}$$

$$= \min \{40 + 0, 6 + 1\} = 7$$

$$dy^{(x)} = \min \{c(y, x) + dx^{(x)}, c(y, z) + dz^{(x)}\}$$

$$= \min \{6 + 0, 1 + 7\} = 6$$

$$dy^{(z)} = \min \{c(y, z) + dz^{(z)}, c(y, x) + dx^{(z)}\}$$

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$$dz^{(x)} = \min \{c(z, x) + dx^{(x)}, c(z, y) + dy^{(x)}\}$$

$$= \min \{40 + 0, 1 + 6\} = 7$$

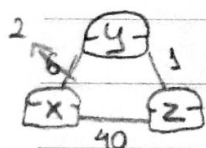
$$dz^{(y)} = \min \{c(z, y) + dy^{(y)}, c(z, x) + dx^{(y)}\}$$

$$= \min \{1 + 0, 40 + 6\} = 1$$

2. x	x y z	x	x y z	x	x y z	x	x y z
x	0 6 7	x	0 6 7	x	0 2 3	x	0 2 3
y	6 0 1	y	6 0 1	y	2 0 1	y	2 0 1
z	7 1 0	z	7 1 0	z	7 1 0	z	3 1 0

y	x y z	y	x y z	y	x y z	y	x y z
x	0 6 7	x	0 6 7	x	0 6 7	x	0 2 3
y	6 0 1	y	2 0 1	y	2 0 1	y	2 0 1
z	7 1 0	z	7 1 0	z	7 1 0	z	3 1 0

z	x y z	z	x y z	z	x y z	z	x y z
x	0 6 7	x	0 6 7	x	0 6 7	x	0 2 3
y	6 0 1	y	6 0 1	y	2 0 1	y	2 0 1
z	7 1 0	z	7 1 0	z	3 1 0	z	3 1 0



y primeiro a descolria

$$d_y(x) = \min \{c(y, x) + d_x^{(x)}, c(y, z) + d_z^{(x)}\}$$

$$= \min \{2 + 0, 1 + 7\} = 2$$

$$d_y(z) = \min \{c(y, x) + d_x^{(z)}, c(y, z) + d_z^{(z)}\}$$

$$= \min \{2 + 7, 1 + 0\} = 1$$

$$d_x^{(y)} = \min \{c(x, y) + d_y^{(y)}, c(x, z) + d_z^{(y)}\}$$

$$= \min \{2 + 0, 40 + 1\} = 2$$

$$d_x^{(z)} = \min \{c(x, y) + d_y^{(z)}, c(x, z) + d_z^{(z)}\}$$

$$= \min \{2 + 1, 40 + 0\} = 3$$

$$d_z^{(x)} = \min \{c(z, x) + d_x^{(x)}, c(z, y) + d_y^{(x)}\}$$

$$= \min \{40 + 0, 1 + 2\} = 3$$

$$d_z^{(y)} = \min \{c(z, x) + d_x^{(y)}, c(z, y) + d_y^{(y)}\}$$

$$= \min \{40 + 2, 1 + 0\} = 1$$

$$d_x^{(y)} = \min \{c(x,y) + d_y^{(y)}, c(x,z) + d_z^{(y)}\}$$

$$\min \{2 + 0, 40 + 1\} = 2$$

$$d_x^{(z)} = \min \{c(x,y) + d_y^{(z)}, c(x,z) + d_z^{(z)}\}$$

$$\min \{2 + 1, 40 + 0\} = 3$$

$$d_y^{(x)} = \min \{c(y,x) + d_x^{(x)}, c(y,z) + d_z^{(x)}\}$$

$$\min \{2 + 0, 1 + 3\} = 2$$

$$d_y^{(z)} = \min \{c(y,x) + d_x^{(z)}, c(y,z) + d_z^{(z)}\}$$

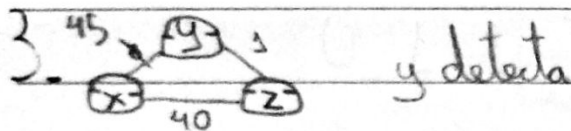
$$\min \{2 + 3, 1 + 0\} = 1$$

$$d_z^{(x)} = \min \{c(z,x) + d_x^{(x)}, c(z,y) + d_y^{(x)}\}$$

$$= \min \{40 + 0, 1 + 2\} = 3$$

$$d_z^{(y)} = \min \{c(z,x) + d_x^{(y)}, c(z,y) + d_y^{(y)}\}$$

$$\min \{40 + 2, 1 + 0\} = 1$$



x	x	y	z	x	x	y	z	x	x	y	z	x	x	y	z
x	0	6	7	x	0	6	7	x	0	4	40	x	0	4	40
y	6	0	1	y	6	0	1	y	8	0	1	y	8	0	1
z	7	1	0	z	7	1	0	z	7	1	0	z	9	1	0

y	x	y	z	y	x	y	z	y	x	y	z	y	x	y	z
x	0	6	7	x	0	6	7	x	0	6	7	x	0	4	40
y	6	0	1	y	8	0	1	y	8	0	1	y	10	0	1
z	7	1	0	z	7	1	0	z	7	1	0	z	9	1	0

z	x	y	z	z	x	y	z	z	x	y	z	z	x	y	z
x	0	6	7	x	0	6	7	x	0	6	7	x	0	4	40
y	6	0	1	y	6	0	1	y	8	0	1	y	8	0	1
z	7	1	0	z	7	1	0	z	9	1	0	z	9	1	0

$$d_y^{(x)} = \min \{c(y,x) + d_x^{(x)}, c(y,z) + d_z^{(x)}\}$$

$$= \min \{45 + 0, 1 + 7\} = 8$$

$$d_y^{(z)} = \min \{c(y,x) + d_x^{(z)}, c(y,z) + d_z^{(z)}\}$$

$$= \min \{45 + 7, 1 + 0\} = 1$$

$$d_x^{(y)} = \min \{c(x,y) + d_y^{(y)}, c(x,z) + d_z^{(y)}\}$$

$$= \min \{45 + 0, 40 + 1\} = 41$$

$$d_x^{(z)} = \min \{c(x,y) + d_y^{(z)}, c(x,z) + d_z^{(z)}\}$$

$$= \min \{45 + 1, 40 + 0\} = 40$$

$$d_z^{(x)} = \min \{c(z,x) + d_x^{(x)}, c(z,y) + d_y^{(x)}\}$$

$$= \min \{40 + 0, 1 + 8\} = 9$$

$$d_z^{(y)} = \min \{c(z,x) + d_x^{(y)}, c(z,y) + d_y^{(y)}\}$$

$$= \min \{40 + 6, 1 + 0\} = 1$$

$$d_x^{(y)} = \min \{c(x,y) + d_y^{(y)}, c(x,z) + d_z^{(y)}\}$$

$$= \min \{45 + 0, 40 + 1\} = 41$$

$$d_x^{(z)} = \min \{c(x,y) + d_y^{(z)}, c(x,z) + d_z^{(z)}\}$$

$$= \min \{45 + 1, 40 + 0\} = 40$$

$$d_y^{(x)} = \min \{c(y,x) + d_x^{(x)}, c(y,z) + d_z^{(x)}\}$$

$$= \min \{45 + 0, 1 + 9\} = 10$$

$$d_y^{(z)} = \min \{c(y,x) + d_x^{(z)}, c(y,z) + d_z^{(z)}\}$$

$$= \min \{45 + 40, 1 + 0\} = 1$$

$$d_z^{(x)} = \min \{c(z,x) + d_x^{(x)}, c(z,y) + d_y^{(x)}\}$$

$$= \min \{40 + 0, 1 + 8\} = 9$$

$$d_z^{(y)} = \min \{c(z,x) + d_x^{(y)}, c(z,y) + d_y^{(y)}\}$$

$$= \min \{40 + 41, 1 + 0\} = 1$$

Se não parar o exercício, potencialmente será feito mais 30 iterações para que se use o caminho entre x e z . Assim chega-se a conclusão que quando o peso aumenta o algoritmo demora para convergir.

4.6

Se a rota de z para chegar a x passa por y , então z deve começar a y que sua distância a x é infinita.

	x	y	z
x	0	6	40
y	∞	∞	∞
z	∞	∞	∞

	x	y	z
x	0	6	7
y	∞	0	1
z	∞	1	0

	x	y	z
x	0	6	7
y	∞	0	1
z	7	1	0

	x	y	z
x	∞	∞	∞
y	6	0	1
z	∞	∞	∞

	x	y	z
x	0	∞	40
y	6	0	1
z	40	∞	0

	x	y	z
x	0	∞	∞
y	6	0	1
z	∞	∞	0

	x	y	z
x	∞	∞	∞
y	∞	∞	∞
z	40	1	0

	x	y	z
x	0	6	∞
y	6	0	∞
z	7	1	0

	x	y	z
x	0	6	7
y	6	0	∞
z	7	1	0

$$d_x^{(1)} = \min \{c(x, y) + d_y^{(1)}, c(x, z) + d_z^{(1)}\}$$

$$= \min \{6 + 0, 40 + 1\} = 6$$

$$d_x^{(2)} = \min \{c(x, y) + d_y^{(2)}, c(x, z) + d_z^{(2)}\}$$

$$= \min \{6 + 1, 40 + 0\} = 7$$

$$d_y^{(1)} = \min \{c(y, x) + d_x^{(1)}, c(y, z) + d_z^{(1)}\}$$

$$= \min \{6 + 0, 1 + 40\} = 6$$

$$d_y^{(2)} = \min \{c(y, x) + d_x^{(2)}, c(y, z) + d_z^{(2)}\}$$

$$= \min \{6 + 40, 1 + 0\} = 1$$

$$d_z^{(1)} = \min \{c(z, x) + d_x^{(1)}, c(z, y) + d_y^{(1)}\}$$

$$= \min \{40 + 0, 1 + 6\} = 7$$

$$d_z^{(2)} = \min \{c(z, x) + d_x^{(2)}, c(z, y) + d_y^{(2)}\}$$

$$= \min \{40 + 6, 1 + 0\} = 1$$

$$dx^{(1)} = \min \{c(x,y) + dy^{(1)}, c(x,z) + dz^{(1)}\}$$

$$= \min \{6 + 0, 40 + 1\} = 6$$

$$dx^{(2)} = \min \{c(x,y) + dy^{(2)}, c(x,z) + dz^{(2)}\}$$

$$= \min \{6 + 1, 40 + 0\} = 7$$

$$dy^{(1)} = \min \{c(y,x) + dx^{(1)}, c(y,z) + dz^{(1)}\}$$

$$= \min \{6 + 0, 1 + \infty\} = 6$$

$$dy^{(2)} = \min \{c(y,x) + dx^{(2)}, c(y,z) + dz^{(2)}\}$$

$$= \min \{6 + \infty, 1 + 0\} = 1$$

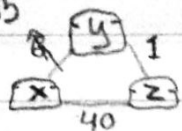
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$$= \min \{40 + 0, 1 + 6\} = 7$$

$$dz^{(2)} = \min \{c(z,x) + dx^{(2)}, c(z,y) + dy^{(2)}\}$$

$$= \min \{40 + 6, 1 + 0\} = 1$$

45



Sendo y o primeiro a perceber

<table><tr><th>x</th><th>x</th><th>y</th><th>z</th></tr><tr><td>x</td><td>0</td><td>6</td><td>7</td></tr><tr><td>y</td><td>∞</td><td>0</td><td>1</td></tr><tr><td>z</td><td>7</td><td>1</td><td>0</td></tr></table>	x	x	y	z	x	0	6	7	y	∞	0	1	z	7	1	0	<table><tr><th>x</th><th>x</th><th>y</th><th>z</th></tr><tr><td>x</td><td>0</td><td>6</td><td>7</td></tr><tr><td>y</td><td>∞</td><td>0</td><td>1</td></tr><tr><td>z</td><td>7</td><td>1</td><td>0</td></tr></table>	x	x	y	z	x	0	6	7	y	∞	0	1	z	7	1	0	<table><tr><th>x</th><th>x</th><th>y</th><th>z</th></tr><tr><td>x</td><td>0</td><td>41</td><td>40</td></tr><tr><td>y</td><td>∞</td><td>0</td><td>1</td></tr><tr><td>z</td><td>7</td><td>1</td><td>0</td></tr></table>	x	x	y	z	x	0	41	40	y	∞	0	1	z	7	1	0	<table><tr><th>x</th><th>x</th><th>y</th><th>z</th></tr><tr><td>x</td><td>0</td><td>41</td><td>40</td></tr><tr><td>y</td><td>∞</td><td>0</td><td>1</td></tr><tr><td>z</td><td>∞</td><td>1</td><td>0</td></tr></table>	x	x	y	z	x	0	41	40	y	∞	0	1	z	∞	1	0	<table><tr><th>x</th><th>x</th><th>y</th><th>z</th></tr><tr><td>x</td><td>0</td><td>41</td><td>40</td></tr><tr><td>y</td><td>41</td><td>0</td><td>1</td></tr><tr><td>z</td><td>∞</td><td>1</td><td>0</td></tr></table>	x	x	y	z	x	0	41	40	y	41	0	1	z	∞	1	0
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$$d_y^{(1)} = \min \{c(y,x) + d_x^{(1)}, c(y,z) + d_z^{(1)}\}$$

$$= \min \{45 + 0, 1 + \infty\} = 45$$

$$d_y^{(2)} = \min \{c(y,x) + d_x^{(2)}, c(y,z) + d_z^{(2)}\}$$

$$= \min \{45 + \infty, 1 + 0\} = 1$$

$$d_x^{(1)} = \min \{c(x,y) + d_y^{(1)}, c(x,z) + d_z^{(1)}\}$$

$$= \min \{45 + 0, 40 + 1\} = 41$$

$$d_x^{(2)} = \min \{c(x,y) + d_y^{(2)}, c(x,z) + d_z^{(2)}\}$$

$$= \min \{45 + 1, 40 + 0\} = 40$$

$$d_z^{(1)} = \min \{c(z,x) + d_x^{(1)}, c(z,y) + d_y^{(1)}\}$$

$$= \min \{40 + 0, 1 + 45\} = 40$$

$$d_z^{(2)} = \min \{c(z,x) + d_x^{(2)}, c(z,y) + d_y^{(2)}\}$$

$$= \min \{40 + 6, 1 + 0\} = 1$$

$$d_x^{(1)} = \min \{c(x,y) + d_y^{(1)}, c(x,z) + d_z^{(1)}\}$$

$$= \min \{45 + 0, 40 + 1\} = 41$$

$$d_x^{(2)} = \min \{c(x,y) + d_y^{(2)}, c(x,z) + d_z^{(2)}\}$$

$$= \min \{45 + 1, 40 + 0\} = 40$$

$$d_y^{(1)} = \min \{c(y,x) + d_x^{(1)}, c(y,z) + d_z^{(1)}\}$$

$$= \min \{45 + 0, 1 + 40\} = 41$$

$$d_y^{(2)} = \min \{c(y,x) + d_x^{(2)}, c(y,z) + d_z^{(2)}\}$$

$$= \min \{45 + 40, 1 + 0\} = 1$$

$$d_z^{(1)} = \min \{c(z,x) + d_x^{(1)}, c(z,y) + d_y^{(1)}\}$$

$$= \min \{40 + 0, 1 + 45\} = 40$$

$$d_z^{(2)} = \min \{c(z,x) + d_x^{(2)}, c(z,y) + d_y^{(2)}\}$$

$$= \min \{40 + 45, 1 + 0\} = 1$$

$$d_x^{(1)} = \min \{c(x,y) + d_y^{(1)}, c(x,z) + d_z^{(1)}\}$$

$$= \min \{45 + 0, 40 + 1\} = 41$$

$$d_x^{(2)} = \min \{c(x,y) + d_y^{(2)}, c(x,z) + d_z^{(2)}\}$$

$$= \min \{45 + 1, 40 + 0\} = 40$$

$$d_z^{(1)} = \min \{40, 1 + \infty\}$$

$$= 40$$

$$d_z^{(2)} = \min \{50 + \infty, 1 + 0\}$$

$$= 1$$

tilibra